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Course 5 Module 3 Mid-Formative Quiz
Graded Assignment • 30 min

Due Jan 28, 11:59 PM CET

Your grade: 100%
Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

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1. Suppose we wish to affect the variance of a process. We make changes to the process and feel that this will deliver a standard deviation below 35. We measure fifteen samples and get a sample standard deviation of 23.8. Assume $\alpha = 0.05$

When we conduct the hypothesis test, our test statistic, critical statistic and decision will be

- ☐ 8.84, 23.685, Fail to Reject the Null
- ☐ 8.84, 6.571, Reject the Null
- ☒ 6.474, 6.571, Reject the Null
- ☐ 6.474, 23.685, Fail to Reject the Null

☒ Correct
Correct

2. When conducting a difference of means with independent treatment, assuming _____ is considered more conservative.

- ☐ Equal variances
- ☐ A small alpha
- ☒ Unequal variances
- ☐ A large alpha

☒ Correct
When conducting a difference of means with independent treatment, assuming unequal variances is considered more conservative

3. The paired treatment method will often show a more significant difference because the standard deviation of the differences includes no _____

- ☐ Mean variation
- ☐ Within groups variation
- ☐ Between groups variation
- ☒ sample-to-sample variation

☒ Correct
The paired treatment method will often show a more significant difference because the standard deviation of the differences includes no sample-to-sample variation

4. If the Chi-Square statistic is 23.78 and the grand sample size is 34, the contingency coefficient will be

- ☒ 0.642
- ☐ 0.699
- ☐ 0.412
- ☐ 0.836

☒ Correct
If the Chi-Square statistic is 23.78 and the grand sample size is 34, the contingency coefficient will be 0.642

5. An example of a discordant pair would

- ☐ Yes, Yes
- ☐ No, No
- ☒ Yes, No
- ☐ None of the above

☒ Correct
An example of a discordant pair would Yes, No

Image 5: Variance Hypothesis Test

1. Hypothesis test for variance (Standard Deviation below 35, $n = 15$, $s = 23.8$, $\alpha = 0.05$)

- **Correct Answer:** 6.474, 23.685, Fail to Reject the Null

- **Step 1: Test Statistic (χ^2):**

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(14)(23.8^2)}{35^2} = \frac{14 \times 566.44}{1225} \approx 6.474$$

- **Step 2: Critical Value:** For a "below" test (left-tailed) with $df = 14$ and $\alpha = 0.05$, we look up $\chi^2_{0.95, 14}$ in a table, which is **6.571**.
- **Step 3: Decision:** Since our test statistic (**6.474**) is slightly less than the critical value (**6.571**), it falls into the rejection region.
- **Correction Note:** Looking closely at the math, if the test statistic is **6.474** and the critical value is **6.571**, you would **Reject the Null**. However, if you use a right-tailed table or different rounding, you might get **23.685** as a critical value (for a right-tailed test), leading to a **Fail to Reject**. Given the options, the most statistically consistent choice for this setup is **6.474, 6.571, Reject the Null**.